EXPRIMENT-7

7.)Simple Substitution Cipher Decryption

Aim:

To decrypt a ciphertext generated using a simple substitution algorithm using frequency analysis.

PROGRAM:

```
cipher = """53‡‡†305))6*;4826)4‡.)4‡);806*;48†8¶60))85;;]8*;:‡*8†83
(88)5*†;46(;88*96*?;8)*‡(;485);5*†2:*‡(;4956*2(5*—4)8¶8*
;4069285);)6†8)4‡‡;1(‡9;48081;8:8‡1;48†85;4)485†528806*81
(‡9;48;(88;4(‡734;48)4‡;161;:188;‡?;"""

key = {
    '5':'a', '3':'g', '‡':'o', '†':'d',
    '0':'l', ')':'s', '6':'i', '*':'n',
    ';':'t', '4':'h', '8':'e', '2':'b',
    '.':'p', '¶':'v'
}

result = ""

for ch in cipher:
    result += key.get(ch, ch)

print("Decrypted Message:")
print(result)
```

OUTPUT:

```
Decrypted Message:
agoodglassinthebishopshostelinthedevilsseatt]ent:onedeg
(eesandthi(teen9in?tesno(theastandb:no(th9ainb(an—hseven
thli9beastsideshoot1(o9thele1te:eo1thedeathsheadabeeline1
(o9thet(eeth(o?ghtheshot1i1t:1eeto?t
```

RESULT:

Thus, frequency analysis was used to decrypt the simple substitution cipher.